

Important Instructions

- This is an **individual assignment**.
 - Copying from colleagues, online repositories, or AI-generated reports without proper modification is considered plagiarism.
 - Any duplicated work will receive **zero penalty** (0 marks).
 - Students must explain and justify all experimental decisions clearly.
 - **Submission Format:** Report + Source Code
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Task 1 — Dataset Selection

Choose a classification dataset

Requirements

- Briefly describe the dataset
 - Mention:
 - Number of samples
 - Number of features
 - Target classes
 - Class distribution
 - Explain why the dataset is suitable for classification
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Task 2 — Baseline Model

Train a baseline machine learning model

Requirements

- Describe preprocessing steps
 - Explain train/test split
 - Mention hyperparameters used
 - Provide implementation screenshots or code snippets
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Task 3 — Evaluation Metrics

Evaluate the model

Requirements

- Compare metric values and interpret them
- Generate and analyze the confusion matrix.
- Discuss what type of errors are most common

Task 4 — Error Analysis

Perform qualitative error analysis on the model predictions.

Requirements

Write at least **three observations** discussing:

- Why the model failed on certain samples

Students should demonstrate critical thinking rather than only reporting metric values.

Task 5 — Experimentation

Design one additional experiment by changing only **one factor**:

Examples include Different hyperparameter, Different preprocessing method, Different classifier, Class balancing technique, ...

Requirements

- Clearly state:
 - Baseline configuration
 - Experimental configuration
 - Compare results fairly
 - Explain whether the change improved performance and why
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Task 6 — Reproducibility

Demonstrate reproducible experimentation practices.

Requirements

Include:

- Random seed usage
- Package versions
- Configuration logging
- Experiment settings

Optional tools:

- MLflow, YAML configuration files
- GitHub repository